ProcureIntel

Procurement intelligence on World Bank open data

A chat-first, AI-assisted platform built as a micro-frontend demo

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Live deployment: https://procurement-intelligence-p7q751lm8-kishans-projects-3ea8dea8.vercel.app/

React 19 · TypeScript · Module Federation · TanStack Query · CopilotKit

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A chat that answers with live cards

Ask in plain words; the assistant draws the lifecycle from stored data

1 Planning
stored: Plans executed

Plans, timeline, country context, plan rows read from World Bank plans

2 Tender design
stored: Tenders published

Published tenders; clauses compared with plans and public contracts

3 Monitoring
stored: Evaluation

Disbursement vs. time, warning signals, schedule Gantt

4 Learning
stored: Feedback

Feedback queue, data readiness, update log

Two modes. Review shows only what is on record for a stored project. Suggest estimates a plan, tender clauses and monitoring for your own new procurement, from medians of similar real projects.

Where users start

Counts come from the stored data



Landing page of the running app

Counted, not typed

9 projects, 605 contract awards and 186 documents, read from the data.

Independent demo

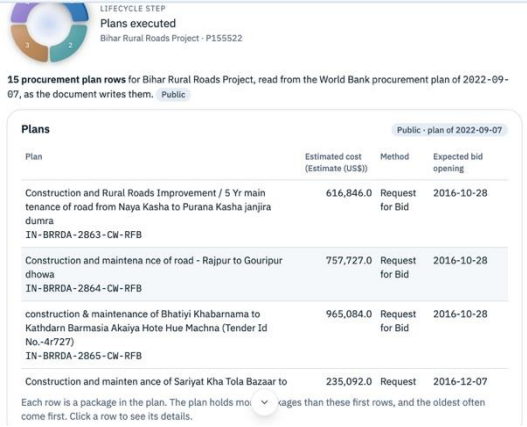
Not an official World Bank product; no World Bank logos.

Review a stored project

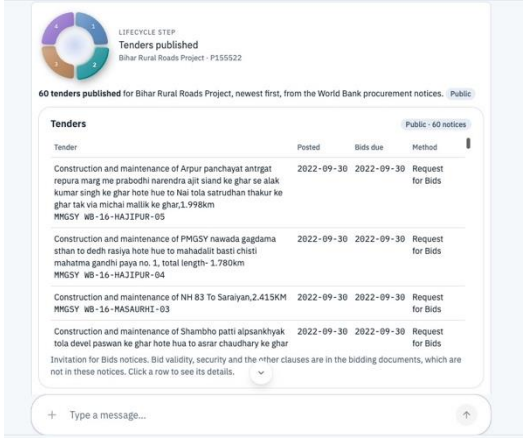
Everything shown is on record; where there is nothing, it says so



The whole lifecycle of a real project



Plans executed: 15 plan rows from the World Bank plan



Tenders published: 60 notices

What is stored today

Counted from the stored data on 2026-11-05

9

road projects

Cambodia, India, Lao PDR, Liberia, Vietnam

605

contract awards

455 suppliers

186

documents

2,445 text chunks

115

published tenders

88 plan rows read from procurement plans

Four public sources

Projects, Contract Awards, Documents and Procurement Notices APIs.

22 assistant tools

Server tools that read this data, and browser tools that draw it.

One host, four remotes, shared singletons

SHELL (host, :5173)

Landing, chat page and its cards, routing, filter rail, theme, the one QueryClient

Planning	Monitoring	Learning	AI
:5174	:5175	:5176	:5177

Remotes expose pages and cards over Module Federation; each sits in its own error boundary and Suspense fallback

Shared singletons: react, react-dom, react-router, react-query, zustand, @pi/data, @pi/events, @pi/config, @pi/ui, @pi/domain

Data API

json-server on :3001 serves the stored data: ingested World Bank snapshots. A reset command rebuilds it.

Assistant :3002

Copilot runtime (AG-UI over HTTP) and ingestion routes. On Vercel the same logic runs as plain JS functions.

A change in one remote shows in all

Four small mechanisms keep the micro-frontends consistent

Server state	**TanStack Query**	One QueryClient in the shell. Every write appends to audit_log and invalidates all queries.
UI state	**Zustand store**	Filter project, fiscal year, role, and chatProjectId: the project the conversation is about, empty until one is shown.
Events	**Typed event bus**	project:selected, fiscalYear:changed, role:changed, enrichment:saved, alert:assigned, feedback:submitted.
Auth seam	**useCurrentUser()**	No real login. All role reads go through one hook, so real auth replaces it without touching components.

Every value says where it came from

Four rules the types and the data layer enforce

Provenance chip on every field

- **Public** — from a public source
- **Entered** — typed in by a user, needs review
- **Derived** — computed, with its sample size
- **Missing** — no source: empty state, action

Never estimated, never filled with an average.

Sample gates, thin-sample flags

Derived results need a minimum n from config. Suggestions under 3 samples are shown and flagged Thin sample.

Each kind of data kept apart

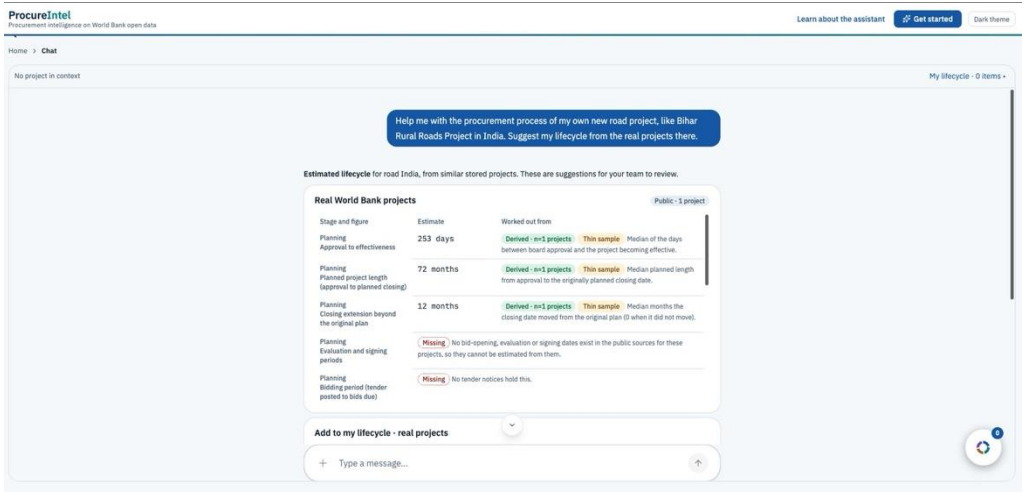
Tools work on a scoped copy of the data, and every group says which kind it is.

Citations, not claims

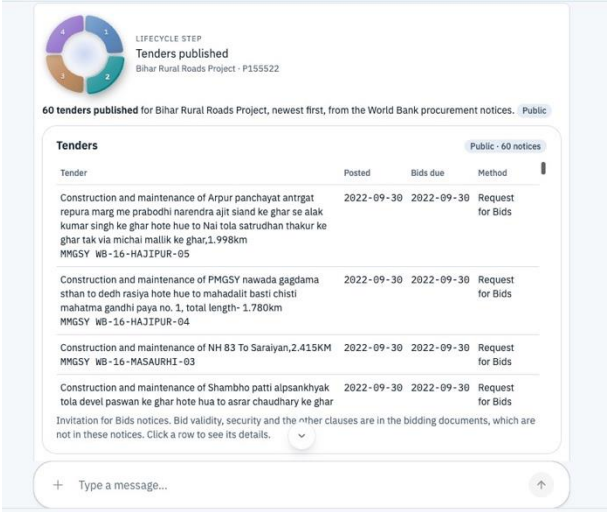
Answers cite record and chunk ids. No support means "No supporting evidence found".

Provenance in practice

Derived with n, Thin sample, Missing and Public chips on live cards



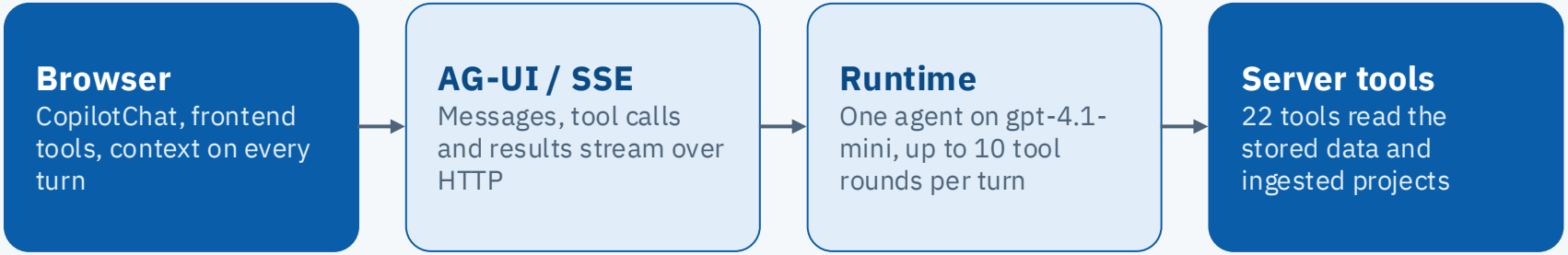
Estimated lifecycle: each figure has its basis, or is Missing



Tenders: Public, from the World Bank notices

The Copilot engine behind the chat

CopilotKit over AG-UI: the model calls tools, the browser draws the result



Three kinds of tool

Server tools look up and compute. Frontend tools run in the page. Render tools draw a result as one of our cards.

Context each turn

The project in conversation, your own lifecycle, and what your own procurement is modelled on.

Fallback

No key, or no answer within 4 s: a rule-based assistant answers from the same data with the same cards.

Rules in writing, certainty in code

Guardrails (A to M)

Show the view before describing it. Keep each kind of data separate. Review vs. suggest. Say "I don't know"; report tool errors and missing values plainly. Keep the project in context. Never claim a view was shown unless a tool drew it.

Analyst standard (N to T)

Understand the context first. Analyse, don't just display. Label documented vs. derived vs. interpretation. Treat gaps as findings with a kind. Argue every comparison. Keep planned, completed, observed and suggested apart.

Models are not deterministic, so code enforces what matters: scoped data per kind, guard() on every tool, the project context cleared after a suggestion, and ingestion that only the user can start.

Citations, or no evidence

Answers cite record and chunk ids; with no support, the assistant says so

Finding: The Bihar Rural Roads Project experienced a 12-month extension of the closing date from December 31, 2022, to December 31, 2023, indicating that the original schedule was not met. The overall risk is rated as Substantial, with fiduciary risk rated Moderate by the World Bank team. These risks are documented as observed in supervision reports and project records, not from planned activities or suggestions.

Evidence: The extension and risk ratings are documented in project records and supervision report 14 [ISR-WB-P155522-14]. Document passages indicate restructuring due to slow progress and disbursement issues [CH-WB-34351121-015, CH-WB-34351121-016]. Other passages note coordination and planning challenges in the sector affecting fund allocation and asset management [CH-WB-34351121-009, CH-WB-34351121-029].

What it means: The extension signals delivery delays, possibly due to slow progress or funding issues, but the exact reasons are not specified in the records. The substantial overall risk suggests significant potential obstacles to project objectives, including procurement and financial management risks.

What to check or do next: Investigate the supervision and completion report texts for explicit reasons behind the extension and risk drivers. Comparing these with similar projects in India, such as the Assam Inland Water Transport Project and the India Second Kerala State Transport Project, may provide insight. The project lacks detailed data on bidders, evaluation outcomes, contract amendments, and lessons extracted from completion reports, limiting deeper risk analysis.

A risk finding for a real project, with report and chunk ids

What do the stored documents say about quantum computing

No supporting evidence was found in the stored documents for the theme of "quantum computing procurement." The app's data does not include references or discussion on this topic in the procurement documents of the stored projects. If you have another topic or a more common procurement theme to explore, please let me know.

A question the documents cannot support

Why it matters

Semantic and keyword search with a similarity floor. Below it, the answer is No supporting evidence found.

From signals to analysis

Rule-based, citing record and chunk ids; no probabilities or scores

Monitoring signals

Rating drop **Risk increase** **Disbursement lag over 20 pts** **Contract past end date** **Amendments**

Analyse

analyse_project, analyse_project_risks, find_comparable_projects and scan_risk_evidence. Priority and comparability rules are stated in the output.

Gaps are findings

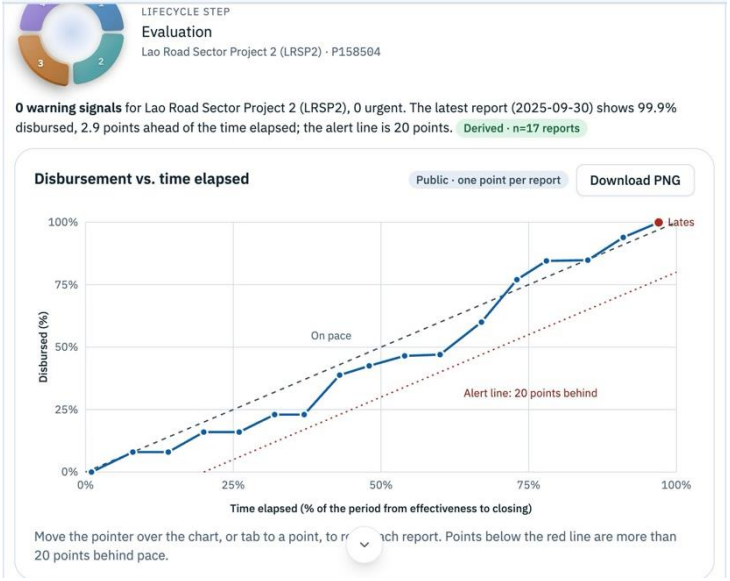
get_data_gaps names a kind for every absence, for example no contract awards for a project, or an unreadable report.

Compare

Bar charts and a scatter plot for 2 to 5 projects. A figure with no source is Missing, never zero. Only comparable projects are compared.

Delivery signals from supervision reports

Disbursement against time elapsed, with the alert line



Evaluation stage of a real project: 17 supervision reports

Signals checked

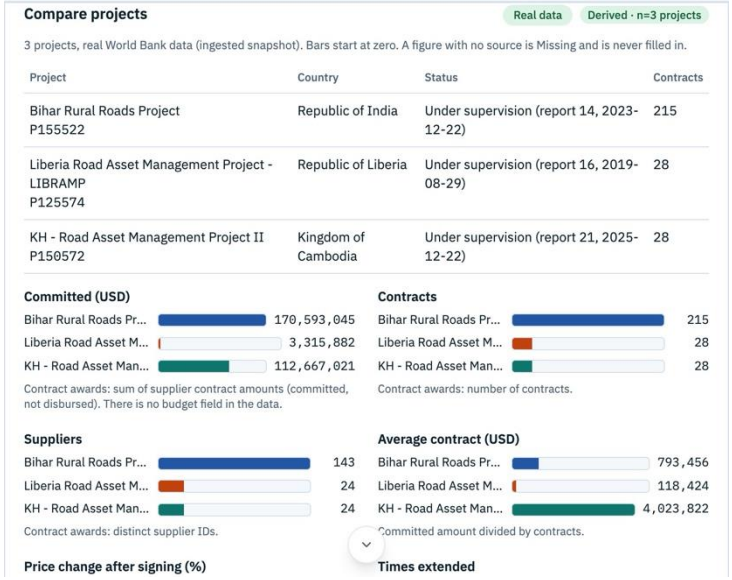
Rating drop, risk increase, disbursement more than 20 points behind time, a contract past its entered end date, approved amendments.

This project

0 warning signals. 99.9% disbursed, 2.9 points ahead of the time elapsed (report of 2025-09-30).

Compare projects side by side

A figure with no source is Missing, never zero



Three real road projects: bars start at zero

How to read it

Bars start at zero. Committed amounts are contract awards, not disbursements, and there is no budget field, so the card says so. Pick any two figures for a scatter plot.

Bring in a project while you watch

One pipeline for the CLI, the local server and the hosted function

find | contracts | notices | documents | texts | parse | plans | chunk | embed | publish

No model | Uses a model

Only the user starts it

find_world_bank_project searches the Projects API; the card offers Proceed. No tool can start an ingestion, so the model cannot.

Safe by design

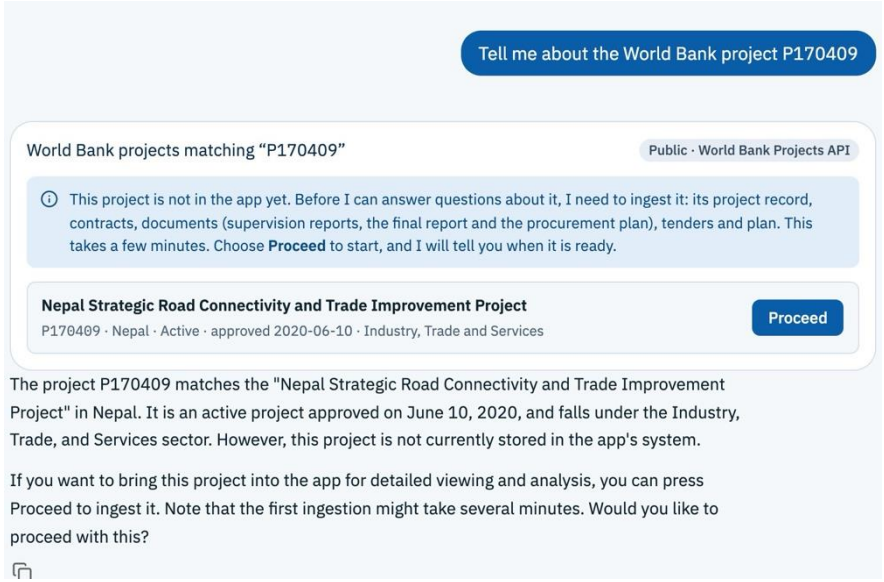
Contact details are never copied. A plan row is kept only if its reference, description and cost appear in the document text. Refused documents are skipped and listed.

Resumable, searchable

Hosted steps pause and continue from a cache. Chunks are embedded, so questions are answered by meaning plus keywords.

The ingest card

A project the app does not hold: found, explained, and only the user presses Proceed



P170409 (Nepal) found through the World Bank Projects API; nothing is ingested yet

What happens next

1. The model searches the Projects API; the card lists the match.
2. Proceed starts the pipeline: record, contracts, documents, tenders, plan, embeddings.
3. An alert and Show details confirm it is ready; the project becomes the one in context.

Help with your own procurement

Suggest mode builds a lifecycle from similar real projects

Review

A stored project: show only what is on record, nothing made up. Past projects are view-only in the chat.

Suggest

propose_lifecycle, plan, tender clauses, suppliers, monitoring and learning options. Each figure is a median with its sample size.

My lifecycle

1 **Planning**

2 **Tender design**

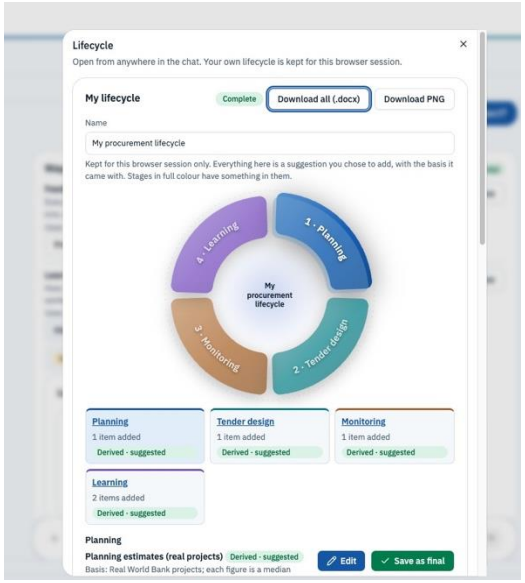
3 **Monitoring**

4 **Learning**

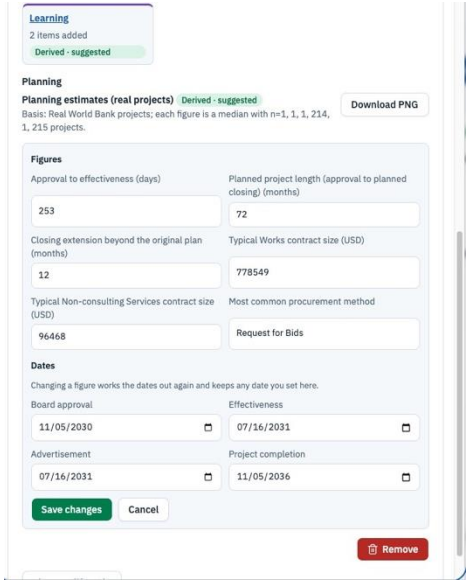
Add from any suggestion card. Edit and Save as final: the item turns from Derived to Entered, keeping its basis. Download each item as PNG, or all of it as Word. Kept in the browser tab only.

Build and export your own lifecycle

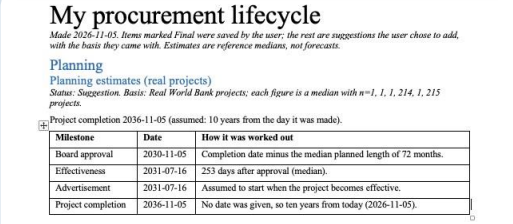
Add suggestions, edit them, then download as PNG or Word



My lifecycle: stages fill in as items are added



Edit figures; dates recalculate



My procurement lifecycle

Made 2026-11-05. Items marked Final were saved by the user; the rest are suggestions the user chose to add, with the basis they came with. Estimates are reference medians, not forecasts.

Planning

Planning estimates (real projects)

Status: Suggestion. Basis: Real World Bank projects; each figure is a median with n=1, 1, 1, 214, 1, 215 projects.

Project completion 2036-11-05 (assumed: 10 years from the day it was made).

Milestone	Date	How it was worked out
Board approval	2030-11-05	Completion date minus the median planned length of 72 months.
Effectiveness	2031-07-16	253 days after approval (median).
Advertisement	2031-07-16	Assumed to start when the project becomes effective.
Project completion	2036-11-05	No date was given, so ten years from today (2026-11-05).

Word export, with the basis of each item

Feedback changes what the platform shows

The continuous-learning loop, end to end

1 → 2 → 3 → 4

1 Helpful / Not helpful

on recommendations and on cards in the chat

2 Feedback record

written to feedback and audit_log

3 Review queue

Reviewer accepts, edits or rejects

4 KPIs and update log

recomputed from the logs

Roles gate who can act

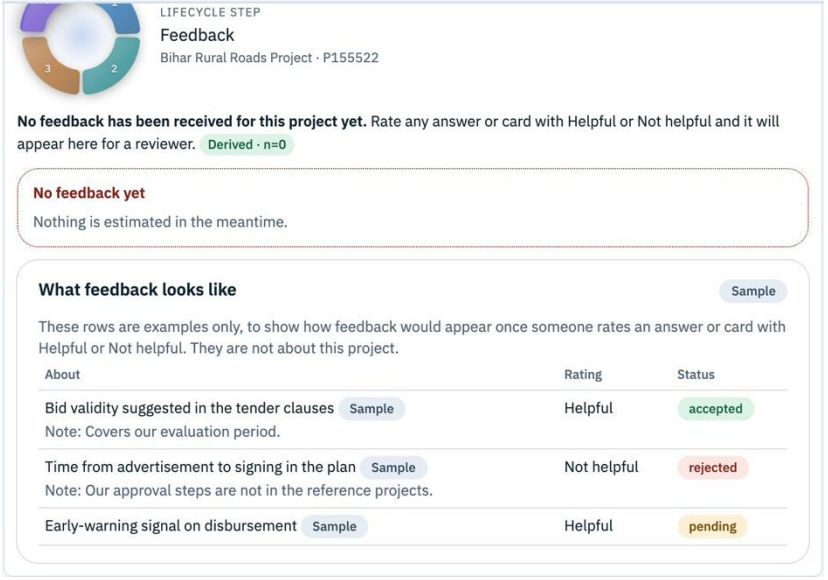
Planner, Contract manager, Reviewer, Admin. A control a role cannot use is disabled with a reason, not hidden.

Honest empty states

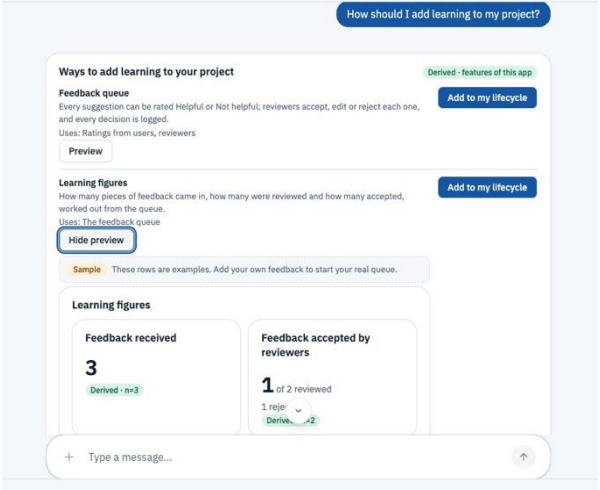
A project with no feedback says so and shows clearly labelled samples with nothing to press.

Learning in the chat

No feedback yet is said plainly; labelled samples show how it will look



Feedback stage of a real project: nothing estimated



Learning options, with a sample preview

Built to keep working during a demo

Failure isolation

RemoteBoundary tells "files could not be fetched" from "loaded then failed". CardBoundary replaces one crashing card with a note and Try again; the chat carries on.

Tests that guard drift

Over 200 Vitest tests (PROGRESS.md). Guards fail on a prompt naming a missing tool, a stale generated function, or a card style regression.

Responsive

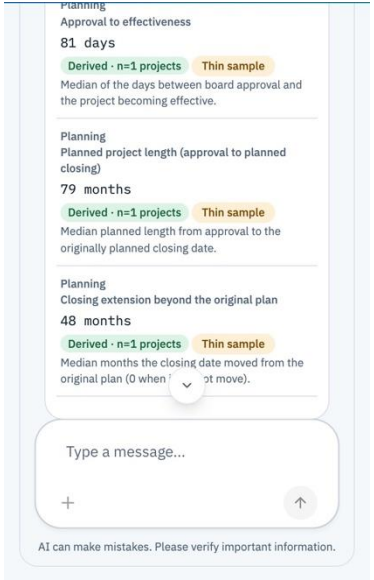
Checked at 375, 768 and desktop widths. Wide tables stack into labelled blocks on a phone; cards use minmax(0, 1fr) so nothing is cut off.

Hosting

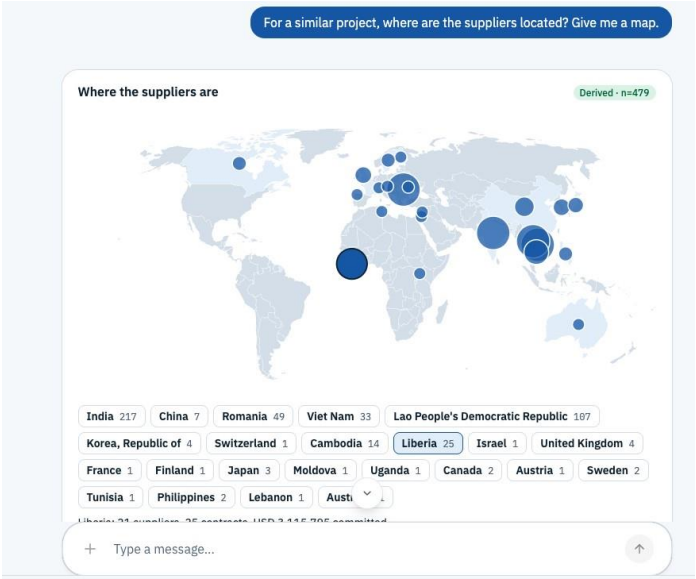
Vercel builds the shell and remotes into one origin. Functions are plain JS or .mjs. Without a key the chat falls back to the rule-based assistant.

Phone-ready, with a supplier map

Cards reflow; tables stack into labelled blocks



Chat card at 390 px wide



Where the suppliers of a similar project are

Layout

Tables stack below 640 px. Checked at 375, 768 and desktop widths.

Assessment dimensions, and where to see them

Dimension	What the UI demonstrates	Status
Ingestion and sources	Live pipeline for World Bank projects, source registry, provenance chips, GPPD context	Built
Knowledge links	Consistent IDs, records linked to document text, citations to chunks	No graph view
Risk models	Monitoring signals and rule-based analysis tools, gaps named by kind	Rules, not trained models
Grounded retrieval	Semantic and keyword search, SourceDrawer, No supporting evidence found	Built
Conflict handling	Public and entered values kept apart, review queue, rule applied and resolution	Built
Continuous learning	Feedback, review queue, derived KPIs, update log	Built
Architecture	Module Federation, shared singletons, package boundaries, ADR 0001	Built

What comes next

Planned steps after this demo

1	**Save My lifecycle beyond the session**	needs a login or a stored collection
2	**Save to Google Docs**	needs Google OAuth
3	**Read more plan tables**	later versions and further rows of each plan
4	**Knowledge-graph view**	records and documents as a graph
5	**Playwright smoke test**	automate the demo path

THANK YOU